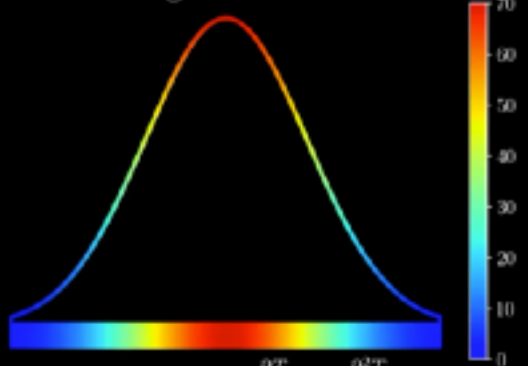


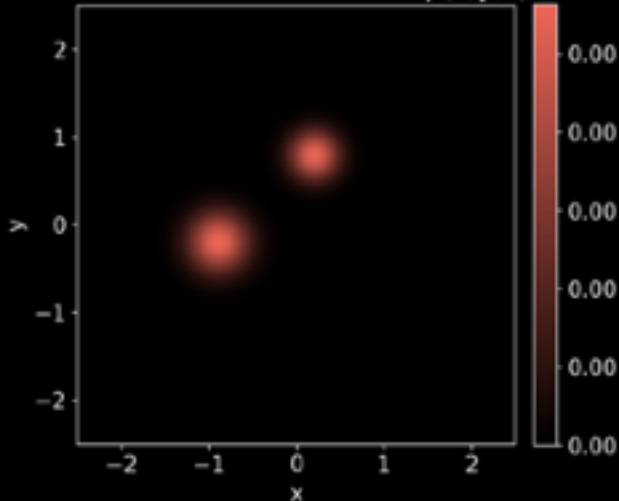
they recover/unify many well-known PDEs

Cooling of a Hot Rod

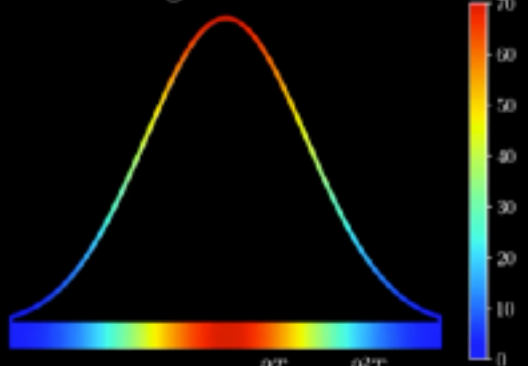


1D Heat Equation: $\frac{\partial T}{\partial t} = \nu \frac{\partial^2 T}{\partial x^2}$ ©RayleighLord

2D Fokker-Planck evolution $\rho(x, y, t)$

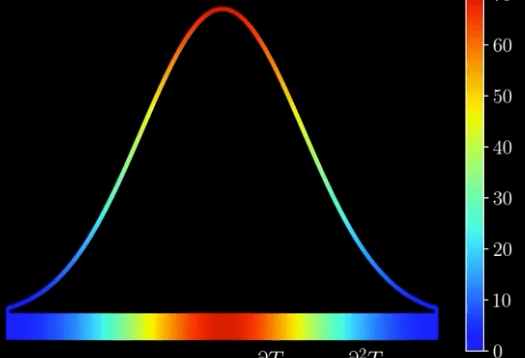


Cooling of a Hot Rod



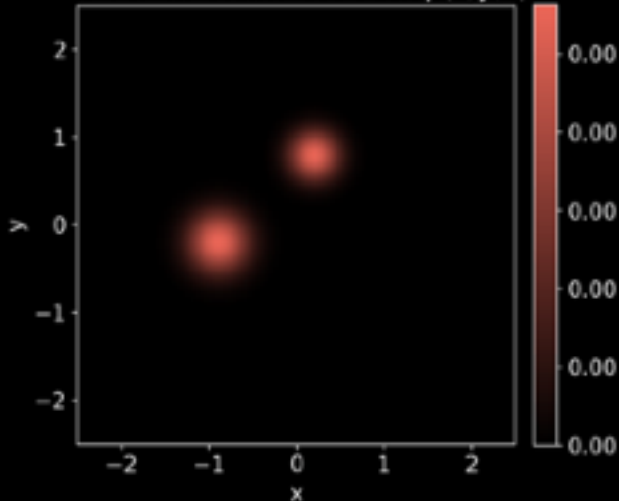
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Cooling of a Hot Rod

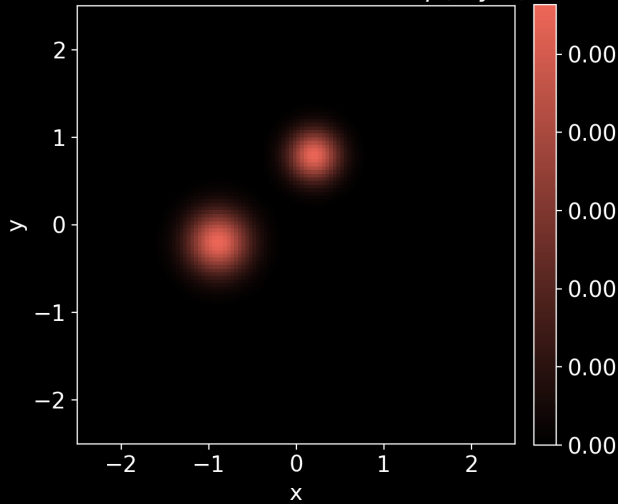


1D Heat Equation: $\frac{\partial T}{\partial t} = \nu \frac{\partial^2 T}{\partial x^2}$ @RayleighLord

2D Fokker-Planck evolution $\rho(x, y, t)$



2D Fokker-Planck evolution $\rho(x, y, t)$



Examples of Energies & PDEs

Key power of Wasserstein grad flows: **they recover/unify many well-known PDEs**

Choice of Energy
Functional $\mathcal{F}$

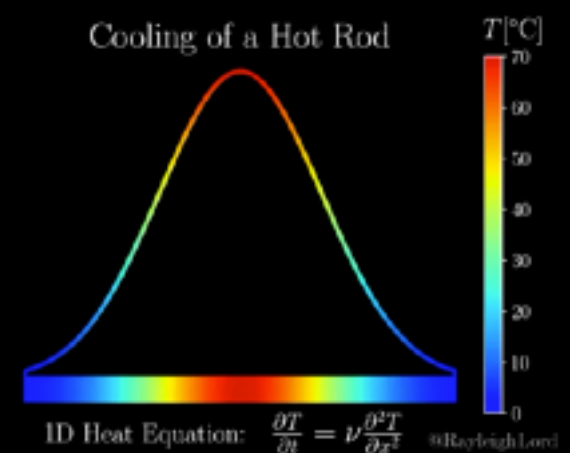
Resulting Gradient Flow
 $\partial_t \rho_t = \nabla_{\mathbf{W}} \mathcal{F}(\rho_t)$

PDE

$$\mathcal{F}(\rho) = \int \rho \log \rho$$

$$\partial_t \rho = \Delta \rho$$

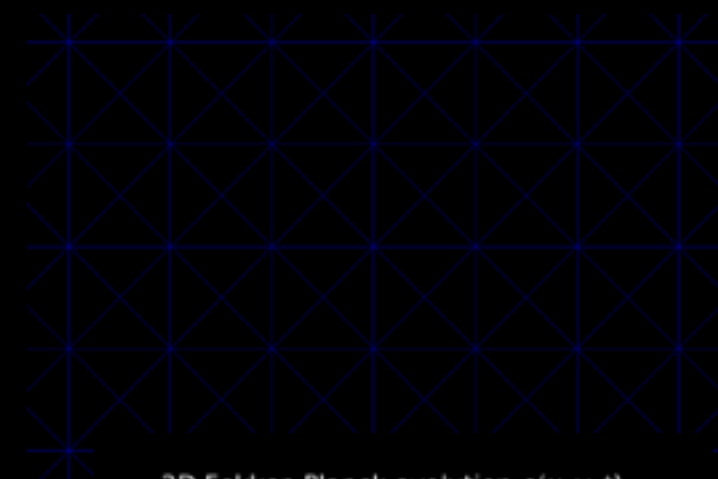
Heat Equation



$$\mathcal{F}(\rho) = \frac{1}{m-1} \int \rho^m$$

$$\partial_t \rho = \Delta(\rho^m)$$

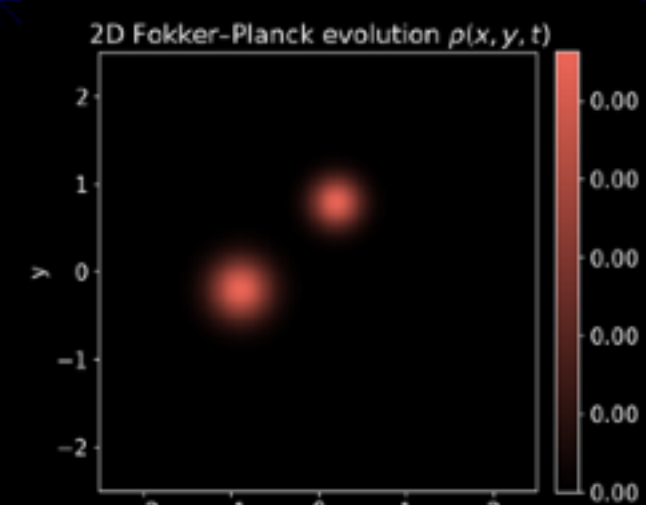
Porous Medium



$$\mathcal{F}(\rho) = \int \rho \log \rho + \int V(x) \rho(x) dx$$

$$\partial_t \rho = \nabla \cdot (\rho \nabla (\log \rho + V))$$

Fokker-Planck



Three Canonical Energies

Three types of “nice” functionals often used in Wasserstein grad flows: